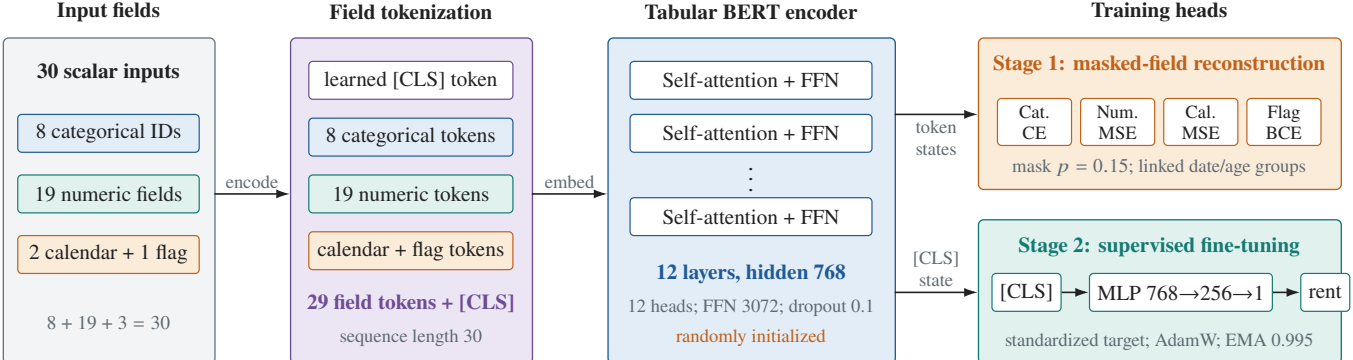

Input fields
30 scalar inputs
8 categorical IDs
19 numeric fields
2 calendar + 1 flag
8 + 19 + 3 = 30
encode
Field tokenization
learned [CLS] token
8 categorical tokens
19 numeric tokens
calendar + flag tokens
29 field tokens + [CLS]
sequence length 30
embed
Tabular BERT encoder
Self-attention + FFN
Self-attention + FFN
Self-attention + FFN
12 layers, hidden 768
12 heads; FFN 3072; dropout 0.1
randomly initialized
token states
[CLS] state
Training heads
Stage 1: masked-field reconstruction
Cat. CE
Num. MSE
Cal. MSE
Flag BCE
mask $p = 0.15$; linked date/age groups
Stage 2: supervised fine-tuning
[CLS]
MLP 768→256→1
rent
standardized target; AdamW; EMA 0.995